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MODULAR FLAT WETBLAST NOZZLE

Abstract

A modular flat wetblast nozzle. The nozzle includes: a slurry inlet chamber coupled to a slurry plate; an air inlet chamber coupled to an air plate; a mixing chamber formed between the slurry plate and air plate; and an air jet manifold removably mountable between the slurry plate and the air plate and configured to receive slurry from the slurry inlet chamber via the slurry plate and to receive air from the air inlet chamber via the air plate, and further configured to force the slurry and air into the mixing chamber to create a flattened mixture that is expelled between the slurry plate and air plate.

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Background/Summary

PRIORITY CLAIM [0001] This application claims priority to U.S. provisional application, 63/562,912, filed on Mar. 8, 2024, Modular Flat Wetblast Nozzle, the contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The subject matter of this invention relates to wetblasting, and more particularly to a modular flat wetblast nozzle that provides uniform surface blasting of a substrate material.

BACKGROUND

[0003] “Wet Blasting” is an industrial process in which pressurized wet slurry is applied to a surface for various cleaning or finishing effects. Various wet-blasting systems exist in which a “wet blast” is jetted onto a workpiece using pressurized air and a slurry formed from a mixture of an abrasive and a liquid. In a typical system, a specially designed, high volume pump mixes abrasive media with water. This slurry mixture is then sent to a nozzle (or nozzles) where regulated compressed air is used to adjust the pressure of the slurry as it blasts the surface. The liquid abrasive effect can be precision engineered to produce desired surface profiles and textures. Unfortunately, controlling the amount of roughness with a high degree of precision in a wetblasting operation in which a media is shot at a surface can be extremely difficult.

SUMMARY

[0004] One aspect of the disclosure provides a modular flat wetblast nozzle, comprising: a slurry inlet chamber coupled to a slurry plate; an air inlet chamber coupled to an air plate; a mixing chamber formed between the slurry plate and air plate; and an air jet manifold removably mountable between the slurry plate and the air plate and configured to receive slurry from the slurry inlet chamber via the slurry plate and to receive air from the air inlet chamber via the air plate, and further configured to force the slurry and air into the mixing chamber to create a flattened mixture that is expelled between the slurry plate and air plate.

[0005] A further aspect of the disclosure includes a robotic wetblasting system having the aforementioned nozzle. In certain aspects, the system may include alternative components, such as different sized manifolds that can be swapped for different applications.

[0006] A further aspect provides a wetblast robot, comprising: a chamber for holding a workpiece; a robotic arm that moves within the chamber; and a flat wetblast nozzle connected to the robotic arm, wherein the flat wetblast nozzle includes: a slurry inlet chamber coupled to a slurry supply hose; an air inlet chamber coupled to an air supply hose; and an air jet manifold configured to receive slurry from the slurry inlet chamber via a slurry plate and to receive air from the air inlet chamber via an air plate, and to force the slurry and air into a mixing chamber to form a flattened mixture that is expelled between the slurry plate and air plate.

[0007] A further aspect of the disclosure provides a method of utilizing the aforementioned nozzle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] These and other features of this invention will be more readily understood from the following detailed description of the various aspects of the invention taken in conjunction with the accompanying drawings in which:

[0009] FIG. 1 shows a modular flat wetblast nozzle according to embodiments.

[0010] FIG. 2 depicts components of the nozzle according to embodiments.

[0011] FIG. 3 depicts further components of the nozzle according to embodiments.

[0012] FIGS. 4a and 4b show the nozzle being deployed in a robotic chamber according to

embodiments.

[0013] FIG. 5 shows a cutaway view showing a bypass fitting according to embodiments.

[0014] The drawings are not necessarily to scale. The drawings are merely schematic representations, not intended to portray specific parameters of the invention. The drawings are intended to depict only typical embodiments of the invention, and therefore should not be considered as limiting the scope of the invention. In the drawings, like numbering represents like elements.

DETAILED DESCRIPTION

[0015] As noted, challenges exist in achieving a precision surface using wetblasting operations. For example, striped irregularities or the like can occur when processing surfaces such as glass, metal, ceramic, plastic, etc. The current approach to overcoming such irregularities when trying to achieve a highly precise and uniform finish is to provide a wide, i.e., flat, nozzle capable of wetblasting a wider path during each pass.

[0016] However, in various applications, a flat wetblast nozzle may need to make several passes when processing larger workpieces. To overcome irregularities such as striping in highly precise applications, the nozzle, slurry and associated settings may need to be fine-tuned using trial and error for the particular type of surface. Accordingly, a one size fits all approach is not practical as, e.g., different blast media sizes may require different nozzle specifications. Furthermore, different parts of the nozzle may wear out over time, requiring costly replacements or rebuilds.

[0017] The present flat wetblast nozzle addresses these challenges by providing a modular design in which components can be easily swapped out to achieve different specifications or to replace worn out parts. The result is a much more robust and cost-effective wetblast solution.

[0018] FIG. 1 depicts a side cut-away view of an illustrative flat wetblast nozzle **10**. Nozzle **10** generally includes: a slurry inlet chamber **12** with an inlet port **13**; a slurry plate **18** that includes a slurry channel **19**; a blast path side fitting **24**; an air plate **20** having an air channel **21** that connects to a compressed air inlet chamber **16**; and an air jet manifold **22** having an slurry feed **39** and an air feed **37** configured to respectively receive slurry from slurry channel **19** and air from air channel **21**. The slurry and air are forced from the manifold **22** and mixed in mixing chamber **23** using beveled surfaces **29** shown on the bottom of the slurry plate **18** and top of the air plate **20**. The resulting mixture is expelled from the mixing chamber **23** to the blast path outlet **14** as a flattened mixture.

[0019] FIG. 2 depicts an exploded isometric view of the slurry inlet chamber **12**, slurry plate **18**, and blast path side fitting **24**. Slurry plate **18** includes a slurry channel **19** that delivers slurry from slurry inlet chamber **12** to the air jet manifold **22** (FIG. 1). As shown, blast path side fitting **24** includes a pair of legs **27** that form a lateral channel that guides the flattened mixture outward between the slurry plate **18** and air plate **20** (not shown in FIG. 2-see FIGS. 1 and 3).

[0020] FIG. 3 depicts an exploded isometric view of the air jet manifold **22**, air plate **20**, and air inlet chamber **16**. With continuing reference to FIG. 1, air jet manifold **22** includes a lateral wall **25** having a plurality of mini-nozzles **31** for forcing air received from the air inlet chamber **16** out into the mixing chamber **23**. As shown in FIG. 1, air jet manifold **22** includes an internal air feed **37** that directs the air to the mini-nozzles **31**. Air jet manifold **22** further includes a partially enclosed surface region **33** with an opening along the lateral wall **25** for forcing slurry received from the slurry inlet chamber **12** into the mixing chamber **23**. Partially enclosed surface region **33** includes a lip **35** along three sides of region **33** that mate with the underside of slurry plate **18**, creating a slurry feed **39** (as shown in FIG. 1) above the mini-nozzles **31**. The air and slurry are expelled from air jet manifold **22** and mixed in the mixing chamber **23** as the two hit the beveled surfaces **29** in air plate **20** and slurry plate **18** (see FIG. 1). The resulting flattened mixture is then expelled to the blast path outlet **14**.

[0021] In one aspect, the air jet manifold is “removably mounted” within the nozzle **10** so that different manifolds **22** can be swapped in/out for different applications. Note that the number and

size of the mini-nozzles (i.e., holes) **31** in a given manifold **22** can be selected and installed for a particular application to achieve a specific result.

[0022] FIGS. **4a** and **4b** depict an image of nozzle **10** in a robotic chamber **40** connected to a slurry supply hose **26** and an air supply hose **28**. As shown, the nozzle **10** is moved by a robot arm **30** and the wetblast is blasted downward onto a work surface from the nozzle **10** to “wide blast” a workpiece (not shown). Robot arm **30** is, e.g., part of an industrial robot that is managed by a set of robot controls.

[0023] FIG. **5** depicts a cut-away view of nozzle **10** cut along 5-5 of FIG. **4b**. Nozzle **10** include a selectable bypass fitting **32** having a bypass or relief gap that helps control the flow of the blast. The size of the gap can be altered by selecting different fittings **32** to help fine tune the flow.

Bypass fitting **32** can be threaded and screwed into a pipe connected to the slurry inlet chamber **12**. [0024] The modular flat wetblast nozzle **10** accordingly takes in slurry (i.e., a mixture of media and water) under pressure from a slurry pump, and then accelerates and atomizes the slurry with compressed air from the air inlet chamber **16** via the air jet manifold **22**. The final mixture is forced out between the slurry plate **18** and the air plate **20** at high velocity with a consistent slurry air mixture distribution. The nozzle **10** can be manufactured in a variety of widths to accommodate different application requirements.

[0025] The nozzle is modular in that individual components such as the air jet manifold **22**, slurry inlet chamber **12**, bypass fittings **32**, and/or other parts can be easily swapped to accommodate, e.g., a different blast media size, blast velocity, coverage requirements, etc. Further, as components wear out, individual components can be easily changed as well. Components may be connected to each other in any manner, e.g., via nuts and bolts, clamps, grooves, etc., and the nozzle **10** can be manufactured from a variety of materials based on weight requirements, durability, lifespan etc.

[0026] The foregoing description of various aspects of the invention has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed, and obviously, many modifications and variations are possible. Such modifications and variations that may be apparent to an individual in the art are included within the scope of the invention as defined by the accompanying claims.

Claims

1. A modular flat wetblast nozzle, comprising: a slurry inlet chamber coupled to a slurry plate; an air inlet chamber coupled to an air plate; a mixing chamber formed between the slurry plate and air plate; and an air jet manifold removably mountable between the slurry plate and the air plate and configured to receive slurry from the slurry inlet chamber via the slurry plate and to receive air from the air inlet chamber via the air plate, and further configured to force the slurry and air into the mixing chamber to create a flattened mixture that is expelled between the slurry plate and air plate.

2. The nozzle of claim 1, further comprising a blast path side fitting positioned around the air jet manifold between the slurry plate and the air plate, wherein the blast path side fitting includes sidewalls configured to direct the flattened mixture out a blast path outlet.

3. The nozzle of claim 2, wherein the air jet manifold includes a lateral wall having a plurality of mini-nozzles for expelling air received from the air inlet chamber into the mixing chamber.

4. The nozzle of claim 3, wherein the air jet manifold further includes a partially enclosed surface region with an opening along the lateral wall for forcing slurry received from the slurry inlet chamber to the mixing chamber.

5. The nozzle of claim 4, wherein the mixing chamber formed adjacent to the lateral wall.

6. The nozzle of claim 5, wherein the slurry plate and air plate include beveled edges within the mixing chamber.

7. The nozzle of claim 3, wherein the air jet manifold further includes an internal air feed that

receives air flow from an air channel in the air plate.

8. The nozzle of claim 4, wherein the partially enclosed surface region includes a lip that mates with the slurry plate about a slurry channel to form a slurry feed.

9. The nozzle of claim 1, wherein the slurry inlet chamber includes a bypass fitting having a relief gap.

10. The nozzle of claim 3, further comprising a second air jet manifold, wherein second air jet manifold includes different sized mini-nozzles than the air jet manifold and is swappable with the air jet manifold.

11. A wetblast robot, comprising: a chamber for holding a workpiece; a robotic arm that moves within the chamber; and a flat wetblast nozzle connected to the robotic arm, wherein the flat wetblast nozzle includes: a slurry inlet chamber coupled to a slurry supply hose; an air inlet chamber coupled to an air supply hose; and an air jet manifold configured to receive slurry from the slurry inlet chamber via a slurry plate and to receive air from the air inlet chamber via an air plate, and to force the slurry and air into a mixing chamber to form a flattened mixture that is expelled between the slurry plate and air plate.

12. The robot of claim 11, further comprising a blast path side fitting positioned around the air jet manifold between the slurry plate and the air plate, wherein the blast path side fitting includes sidewalls configured to direct the flattened mixture out a blast path outlet.

13. The robot of claim 12, wherein the air jet manifold includes a lateral wall having a plurality of mini-nozzles for forcing air received from the air inlet chamber into the mixing chamber.

14. The robot of claim 13, wherein the air jet manifold further includes a partially enclosed surface region with an opening along the lateral wall for forcing slurry received from the slurry inlet chamber into the mixing chamber.

15. The robot of claim 14, wherein the slurry plate and air plate form the mixing chamber adjacent to the lateral wall.

16. The robot of claim 15, wherein the slurry plate and air plate include beveled edges within the mixing chamber.

17. The robot of claim 16, wherein the air jet manifold further includes an internal air feed that directs air flow from an air channel in the air plate to the mini-nozzles.

18. The robot of claim 14, wherein the partially enclosed surface region includes a lip that mates with the slurry plate about a slurry channel to form a slurry feed.

19. The robot of claim 11, wherein the slurry inlet chamber includes a bypass fitting having a relief gap.

20. The robot of claim 13, further comprising a second air jet manifold, wherein second air jet manifold includes different sized mini-nozzles than the air jet manifold and is swappable with the air jet manifold.
